

The axioms of n -exact categories

BACHELOR'S THESIS

THOMAS ROSSET

Norwegian University of Science and Technology, Trondheim
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Johanne Haugland

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Abstract

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Introduction

0.1 Prerequisites

In order to understand this paper, the reader should have some knowledge about modern homological algebra. In particular, they should already be familiar with category theory, more specifically abelian category theory, as is presented for example in the lecture notes of MA3204 Homological Algebra [\[5\]](#).

Conventions and notations

Chapter 1

An introduction to exact categories

As a way to generalize the notion of abelian categories, one can look at a general additive category on which we define a certain notion of short exact sequences that has very similar properties to the abelian category counterpart. This chapter aims to motivate the definition of these so-called exact categories, to later define them formally and generalize some classical results from abelian categories.

In this chapter, the word kernel (respectively cokernel) will usually refer to the kernel morphism (respectively cokernel morphism), not the object. The content of this chapter is inspired by both the lecture notes of MA3204 Homological Algebra [5] and the article about exact categories by Theo Bühler [2].

1.1 Motivation

When studying abelian categories, there are many convenient properties related to monics and epics, which are used to prove results about short exact sequences. Let us recall a few:

Proposition 1.1 *In an abelian category,*

- *every monic is a kernel of its cokernel;*
- *every epic is a cokernel of its kernel.*

Furthermore, the following properties hold:

- *the pushout of a monic is a monic;*
- *the pullback of a monic is a monic;*
- *the pushout of an epic is an epic;*
- *the pullback of an epic is an epic.*

We refer the reader to [5, Sections 12-13] for the proofs of these facts.

Fundamentally, this tells us that every monic and every epic fits into a short exact sequence, and that taking the pullback or pushout of a morphism that is part of a short exact sequence gives rise to another short exact sequence. Indeed, if $f : A \rightarrow B$ is monic, then it fits into the short exact sequence $A \xrightarrow{f} B \xrightarrow{\text{cok}f} \text{Cok } f$. Furthermore, since morphisms that are part of a short exact sequence are either monic or epic, they will once again be monic or epic after a pullback or pushout, thus will once again fit in a short exact sequence.

Let us quickly recall what it means for a class of composable morphisms to be closed under isomorphisms.

Definition 1.2 Let $\mathcal{E}$ be a class of composable morphisms, i.e. a set of pairs (f, g) where the codomain of f is the domain of g . We say $\mathcal{E}$ is

closed under isomorphisms if, for any commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow a & & \downarrow b & & \downarrow c \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$$

where $(f, g) \in \mathcal{E}$ and the vertical maps are isomorphisms, $(f', g') \in \mathcal{E}$.

There are many properties we can conclude about kernels and cokernels using the previous results and this definition. These properties are what will motivate the definition of exact categories.

Proposition 1.3 *Let $\mathcal{A}$ be an abelian category and let $\mathcal{E}$ be the class of short exact sequences. $\mathcal{E}$ is closed under isomorphisms. Furthermore, the following hold:*

- [E0] *For any object $A \in \mathcal{A}$, the identity morphism 1_A is a kernel;*
- [E0^{op}] *For any object $A \in \mathcal{A}$, the identity morphism 1_A is a cokernel;*
- [E1] *The class of kernels is closed under composition;*
- [E1^{op}] *The class of cokernels is closed under composition;*
- [E2] *The pushout of a kernel along any morphism is a kernel;*
- [E2^{op}] *The pullback of a cokernel along any morphism is a cokernel.*

Proof It is clear that the class of short exact sequences is closed under isomorphisms. Indeed, $A \xrightarrow{f} B \xrightarrow{g} C$ is a short exact sequence if and only if f is a kernel of g and g is a cokernel of f . The result follows from the fact that kernels and cokernels are preserved by isomorphisms.

[E0] and [E0^{op}] follow directly from the fact that the identity morphism is both the kernel and cokernel of the zero map.

We can deduce [E1] and [E1^{op}] by recalling that composing monics (respectively epics) gives us monics and that monics are kernels of their cokernel (respectively epics are cokernels of their kernel).

As for [E2], recall that kernels are always monic. Thus, by 1.1, the pushout of a kernel is monic as well. But then, by 1.1 again, it is a kernel of its cokernel, thus it is a kernel. The proof for $[E2^{op}]$ is similar. $\square$

Remark 1.5 We can in a very similar way prove that the pullback of a kernel (respectively the pushout of a cokernel) is a kernel (respectively cokernel), however we choose not to include this as it turns out to be unnecessary for many results.

1.2 Exact Categories

Based on the observations from the last section, one can define a more general notion that encapsulates the idea of short exact sequences, namely exact categories. To do so, let us first define kernel-cokernel pairs.

Definition 1.6 Let $\mathcal{A}$ be an additive category. A *kernel-cokernel pair* (i, p) in $\mathcal{A}$ is a pair of composable morphisms $A \xrightarrow{i} B \xrightarrow{p} C$ such that i is a kernel of p and p is a cokernel of i .

In the context of abelian categories, we notice that kernel-cokernel pairs indeed give rise to short exact sequences in the usual sense. In a general additive category, however, we cannot say the same since we have no guarantee that kernels and cokernels even exist for arbitrary morphisms, and thus we are unsure that $\text{Im } i$ exists. But we will use our intuition from abelian categories to expand the notion of short exact sequences to additive categories. First, let us fix some terminology.

Definition 1.7 Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$. We call a morphism $i : A \rightarrow B$ an *inflation*

(or *admissible monic*) if there exists a morphism $p : B \rightarrow C$ such that $(i, p) \in \mathcal{E}$. Dually, we call a morphism $q : B \rightarrow C$ a *deflation* (or *admissible epic*) if there exists a morphism $j : A \rightarrow B$ such that $(j, q) \in \mathcal{E}$. We use $A \xrightarrow{i} B$ for inflations, and $B \xrightarrow{p} C$ for deflations. We say $\mathcal{E}$ is *closed under isomorphisms* if whenever we have $A \xrightarrow{i} B \xrightarrow{p} C \in \mathcal{E}$ fitting in a commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ \downarrow a & & \downarrow b & & \downarrow c \\ A' & \xrightarrow{j} & B' & \xrightarrow{q} & C' \end{array}$$

with a , b , and c isomorphisms, then $(j, q) \in \mathcal{E}$.

It is usually convenient to require being closed under isomorphisms, as objects with universal properties are generally unique up to isomorphism. In particular, recall that all kernel objects of a specific morphism are isomorphic. In the case of a class of kernel-cokernel pairs $\mathcal{E}$, being closed under isomorphisms implies that all kernels of a deflation are inflations. Indeed, consider $B \xrightarrow{p} C$ a deflation and $A \xrightarrow{i} B$ an inflation such that $(i, p) \in \mathcal{E}$. Let $A' \xrightarrow{j} B$ be another kernel of p and $A \xrightarrow{a} A'$ be the isomorphism between both kernels. The sequence $A' \xrightarrow{j} B \xrightarrow{p} C$ fits in the commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ \downarrow a & & \downarrow 1_B & & \downarrow 1_C \\ A' & \xrightarrow{j} & B & \xrightarrow{p} & C \end{array}$$

Since all vertical arrows are isomorphisms, we conclude that $(j, p) \in \mathcal{E}$. The proof that all cokernels of an inflation are deflations is dual.

We can now proceed to define exact categories, basing our definition on the properties we established for abelian categories.

Definition 1.8 Let $\mathcal{A}$ be an additive category. An *exact structure* on $\mathcal{A}$ is a class $\mathcal{E}$ of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphisms and satisfies the following axioms:

- [E0] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an inflation;
- [E0^{op}] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is a deflation;
- [E1] The composition of two inflations yields an inflation;
- [E1^{op}] The composition of two deflations yields a deflation;
- [E2] The pushout of an inflation along any morphism yields an inflation;
- [E2^{op}] The pullback of a deflation along any morphism yields a deflation.

The last two axioms can be summarized by the following commutative diagrams:

$$\begin{array}{ccc}
 C & \xrightarrow{f'} & D \\
 c \downarrow & \text{PO} & \downarrow d \\
 C' & \xrightarrow{\quad f \quad} & D'
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 C' & \overset{f'}{\dashrightarrow} & D' \\
 c \downarrow & \text{PB} & \downarrow d \\
 C & \xrightarrow{\quad f \quad} & D
 \end{array}$$

If $\mathcal{E}$ is an exact structure on $\mathcal{A}$, we say $(\mathcal{A}, \mathcal{E})$ is an *exact category* and the elements of $\mathcal{E}$ are called *short exact sequences* (or *conflations*).

1.3 Extension-closed subcategories

In 1.1, we saw that all abelian categories are exact categories, with a possible exact structure being all short exact sequences. We can generalize this example of exact categories to extension-closed subcategories of abelian categories. Let us first clarify what this means.

Definition 1.9 Let $\mathcal{A}$ be an abelian category and $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory of $\mathcal{A}$. $\mathcal{C}$ is said to be *closed under extension* if, for any short exact sequence in $\mathcal{A}$

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

such that A and C are both in $\mathcal{C}$, then B is also in $\mathcal{C}$.

In that context, we say that B is an *extension of C by A* .

Proposition 1.10 *Let $\mathcal{A}$ be an abelian category and $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory of $\mathcal{A}$ that is closed under extensions. Let $\mathcal{E}$ be the collection of short exact sequences in $\mathcal{A}$ such that all three objects are in $\mathcal{C}$. Then $(\mathcal{C}, \mathcal{E})$ is an exact category.*

Proof We prove each of the axioms one by one.

$\mathcal{E}$ is closed under isomorphisms since short exact sequences in $\mathcal{A}$ are closed under isomorphisms.

Since $\mathcal{C}$ is an additive subcategory, it contains the zero object. As such, for any object $B \in \mathcal{C}$, the sequences $0 \rightarrow B \xrightarrow{1_B} B$ and $B \xrightarrow{1_B} B \rightarrow 0$ are short exact and contained in $\mathcal{C}$, i.e. identity morphisms are both deflations and inflations.

Now let us consider the axiom [E2]. Let $X \xrightarrow{i} Y \xrightarrow{d} Z$ be a short exact sequence in $\mathcal{C}$ and $f : A \rightarrow X$. We can consider the pushout diagram in the ambient category $\mathcal{A}$:

$$\begin{array}{ccccc} X & \xrightarrow{i} & Y & \xrightarrow{d} & Z \\ f \downarrow & & \downarrow f' & & \downarrow f'' \\ A & \xrightarrow{j} & B & \xrightarrow{e} & C \end{array}$$

where $C = \text{Cok } j$. From basic results in abelian categories, we know that j is monic, and that we can choose $C = Z$ (see [5, Proposition 13.7] for the latter). Thus the sequence $A \xrightarrow{j} B \xrightarrow{e} Z$ is short exact, with both A and Z in $\mathcal{C}$, which implies that $B \in \mathcal{C}$. We therefore get a short exact sequence in $\mathcal{C}$, and j is an inflation. The proof for [E2^{op}] is dual.

To prove [E1], let us consider short exact sequences $A \xrightarrow{i} B \xrightarrow{d} C$ and $B \xrightarrow{j} Y \xrightarrow{e} Z$ in $\mathcal{C}$. Then we can construct the following pushout in the ambient category:

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{j} & Y \\ & & \downarrow d & \searrow r & \downarrow f \\ & & C & \xrightarrow{k} & E \end{array}$$

By 1.1, we know that f is epic. It then remains to prove that $j \circ i$ is a kernel of f . Using [5, Corollary 13.6], we know the square is also a pullback since i is monic. Consider $g : X \rightarrow Y$ such that $f \circ g = 0$. Using the pullback property on $g : X \rightarrow Y$ and $0 : X \rightarrow C$, we get a unique map $\hat{g} : X \rightarrow B$ such that $j \circ \hat{g} = g$ and $d \circ \hat{g} = 0$. Now using that i is a kernel of d , we get a unique map $\hat{\hat{g}} : X \rightarrow A$ such that $i \circ \hat{\hat{g}} = \hat{g}$. In particular, $(j \circ i) \circ \hat{\hat{g}} = g$, and $\hat{\hat{g}}$ is the unique map satisfying this property. Finally, we note that

$$f \circ (j \circ i) = (f \circ j) \circ i = (k \circ d) \circ i = k \circ (d \circ i) = 0$$

so that we can conclude that $j \circ i$ is a kernel of f . Since f is epic and $j \circ i$ is its kernel, we conclude that $A \xrightarrow{j \circ i} Y \xrightarrow{f} E$ is a short exact sequence in $\mathcal{C}$. As such, inflations are closed under composition. Proving [E1^{op}] is dual.

□

Remark 1.12 A slightly modified converse statement is also true, and is known as the *Quillen-Gabriel embedding theorem*. It states that, if $(\mathcal{A}, \mathcal{E})$ is an exact category where $\mathcal{A}$ is small, then there is an embedding $\mathcal{A} \hookrightarrow \mathcal{B}$ such that $\mathcal{B}$ is an abelian category and $\mathcal{E}$ is the class of all short exact sequences in $\mathcal{B}$.

1.4 Examples

The previous statement is quite useful for finding exact categories that are not abelian categories. Here are a few fundamental general examples.

Examples 1.13

- Let R be a non-noetherian unital ring. Consider the category of finitely generated R -modules $\text{mod}R$ (seen as a subcategory of $\text{Mod}R$). It is well known that this category is closed under extensions. However, it is not abelian. Indeed, consider a non-finitely generated ideal $I \subseteq R$ (which exists since R is non-noetherian). Then the kernel of the quotient map $R \rightarrow R/I$ is the ideal I . In that case, both R and R/I are finitely generated R -modules, but I is by definition not finitely generated. Thus $\text{mod}R$ does not contain the kernel of all its maps, and cannot be abelian;
- Let R be any ring. Consider the full subcategory of $\text{mod}R$ consisting of all projectives $\mathcal{P}(R)$. Let $P, P' \in \mathcal{P}(R)$ be two projective R -modules. It is well known from homological algebra that $\text{Ext}^1(P', P) = 0$. But then, considering Ext^1 as the Yoneda extension functor, this implies that any short exact sequence $0 \rightarrow P \rightarrow E \rightarrow P' \rightarrow 0$ splits, i.e. $E = P \oplus P'$. Since the direct sum of two projectives is projective, this shows that $\mathcal{P}(R)$ is closed under extensions. However, the cokernel of the map $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is $\mathbb{Z}/2\mathbb{Z}$, which is not projective, so it is not abelian.

1.5 Reducing the axioms

As was shown by Bernhard Keller in [4, Appendix A], some of the axioms required in the classical definition of exact categories are redundant.

Proposition 1.14 *Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphisms. Assume that $\mathcal{E}$ satisfies $[\text{E1}^{op}]$, $[\text{E2}]$ as well as*

- $[\text{E0}']$ 1_0 is a deflation;
 $[\text{E2}^{op}']$ For any $f : C \rightarrow Z$ and deflation $d : Y \twoheadrightarrow Z$, there is a commutative square

$$\begin{array}{ccc} Y' & \xrightarrow{d'} \twoheadrightarrow & Z' \\ f' \downarrow & & \downarrow f \\ Y & \xrightarrow{d} \twoheadrightarrow & Z \end{array}$$

Then $\mathcal{E}$ is an exact structure on $\mathcal{E}$.

Proof We will first prove that $[\text{E2}]$ and $[\text{E2}^{op}']$ imply $[\text{E2}^{op}]$ by explicitly building a pullback. Consider the commutative diagram from $[\text{E2}^{op}']$:

$$\begin{array}{ccccc} K & \xrightarrow{i'} & Y' & \xrightarrow{d'} \twoheadrightarrow & Z' \\ f_1 \downarrow & & f_2 \downarrow & & f_3 \downarrow \\ X & \xrightarrow{i} & Y & \xrightarrow{d} \twoheadrightarrow & Z \end{array}$$

f_1 exists as the map between kernels. As such we can build the pushout of i' along f_1 , which we will name P . This will be the pullback of d along f . To prove that, construct the maps $a : P \rightarrow Z'$ and $b : P \xrightarrow{Y}$ by applying the pushout property to the following commutative diagrams:

$$\begin{array}{ccc} \begin{array}{ccccc} K & \xrightarrow{i'} & Y' & & \\ f_1 \downarrow & & \downarrow \hat{f}_1 & & \\ X & \xrightarrow{i'} & P & \xrightarrow{d} & Z' \\ & \searrow \hat{i}' & \dashrightarrow \exists! a & & \\ & 0 & \searrow & & \end{array} & \text{and} & \begin{array}{ccccc} K & \xrightarrow{i'} & Y' & & \\ f_1 \downarrow & & \downarrow \hat{f}_1 & & \\ X & \xrightarrow{i'} & P & \xrightarrow{f_2} & Y \\ & \searrow \hat{i}' & \dashrightarrow \exists! b & & \\ & i & \searrow & & \end{array} \end{array}$$

For P to be a pullback, it is necessary that $d \circ b = f_3 \circ a$ (since a pullback gives a commutative square). But this is true as both maps are a solution to the pushout problem

$$\begin{array}{ccccc}
 K & \xrightarrow{i'} & Y' & \xrightarrow{d'} & Z' \\
 f_1 \downarrow & & \downarrow \hat{f}_1 & & \downarrow \\
 X & \xrightarrow{\hat{i}'} & P & & f_3 \\
 \downarrow i & & \searrow d = f_3 \circ a & & \downarrow \\
 Y & \xrightarrow{\quad d \quad} & & \searrow a & Z
 \end{array}$$

Clearly $(d \circ i) \circ f_1 = 0 = f_3 \circ (d' \circ i')$, so that the big square commutes. Thus there exists a unique map $p : P \rightarrow Z$ such that $p \circ \hat{f}_1 = f_3 \circ d'$ and $p \circ \hat{i}' = d \circ i = 0$. But note that $f_3 \circ (a \circ \hat{f}_1) = f_3 \circ d'$ and $f_3 \circ (a \circ \hat{i}') = 0$ by definition of a , so $f_3 \circ a = p$. Similarly, $d \circ (b \circ \hat{f}_1) = d \circ f_2 = f_3 \circ d'$ by commutativity and $d \circ (b \circ \hat{i}') = d \circ i = 0$, so $d \circ b = p = f_3 \circ a$.

That [E0'] implies both [E0] and [E0^{op}] is pretty straightforward. $\square$

Chapter 2

Expanding to n -exact categories

After generalizing from abelian categories to exact categories, we can go one step further by looking at longer exact sequences. First, we will redefine the notions needed for exact categories, but in a more general context. Then we will proceed to define n -exact categories, and give some concrete example of such categories.

When not stated otherwise, $\mathcal{A}$ denotes an additive category. The contents of this chapter are based on [3].

2.1 Working in higher degrees

Before defining n -exact categories, there are many prerequisites that need to be addressed. Firstly, we will need a way to properly describe what the equivalent of a kernel-cokernel pair would be in this new

context. As a result, it makes sense to define n -kernels and n -cokernels, which require the notion of weak kernel (respectively weak cokernel).

Definition 2.1 Let $A \xrightarrow{f} B$ be a morphism in $\mathcal{A}$. A *weak kernel* of f is an object $K \in \mathcal{A}$ with a morphism $K \xrightarrow{k} A$ such that the sequence of abelian groups

$$\mathcal{A}(C', K) \xrightarrow{k \circ -} \mathcal{A}(C', A) \xrightarrow{f \circ -} \mathcal{A}(C', B)$$

is exact for all $C' \in \mathcal{A}$.

Dually, a *weak cokernel* of f is an object $C \in \mathcal{A}$ with a morphism $B \xrightarrow{g} C$ such that the sequence of abelian groups

$$\mathcal{A}(C, C') \xrightarrow{- \circ k} \mathcal{A}(B, C') \xrightarrow{- \circ f} \mathcal{A}(A, C')$$

is exact for all $C' \in \mathcal{A}$.

Remarks 2.2 We do not ask for the sequence to be short exact, but only for it to be exact in the middle term. Furthermore, the following observations are quite useful in practice:

- (a) Notice the direction of the sequences is different since for weak cokernels, we apply the contravariant functor $\mathcal{A}(-, C')$ to our original sequence, whereas we apply the covariant functor $\mathcal{A}(C', -)$ in the case of weak kernels.
- (b) For $g : K \rightarrow A$ a weak kernel of $f : A \rightarrow B$, the sequence being exact lets us know that
 - (i) The composition $f \cdot g$ is 0;
 - (ii) For any morphism $h : C \rightarrow A$ such that $f \cdot h = 0$, there exists a morphism $\hat{h} : C \rightarrow K$ such that $g \circ \hat{h} = h$.

That last point can be summarized by the following diagram:

$$\begin{array}{ccccc}
 K & \xrightarrow{g} & A & \xrightarrow{f} & B \\
 \uparrow \exists \hat{h} & & \nearrow h & \searrow 0 & \\
 C & & & &
 \end{array}$$

These two points follow from exactness in the middle term. Indeed, since $\text{Im } (k \circ -) = \text{Ker } (f \circ -)$, any map $h : C \rightarrow A$ such that $f \circ h = 0$, is in the image of the post-composition with k , i.e. can be factored (not necessarily uniquely!) as $k \circ \hat{h}$. In particular, if we consider $C' = K$, we conclude that

$$k = k \circ 1_K \in \text{Im } (k \circ -) = \text{Ker } (f \circ -),$$

i.e. $f \circ k = 0$. The result for weak cokernels is dual.

In summary, weak cokernels and weak kernels have a very similar property to that of cokernels and kernels, except that uniqueness in our factoring of morphisms is not guaranteed. In other words, the only difference is that a weak kernel is not necessarily monic, and a weak cokernel is not necessarily epic.

This notion is quite fundamental in the elaboration of n -kernels and n -cokernels:

Definition 2.3 Let $X^0 \xrightarrow{d^0} X^1$ be a morphism in $\mathcal{A}$. An n -kernel of d^0 is a sequence

$$X^{-n} \xrightarrow{d^{-n}} X^{-n+1} \xrightarrow{d^{-n+1}} \dots \xrightarrow{d^0} X^0$$

such that the sequence induced by applying the functor $\mathcal{A}(C', -)$

$$0 \longrightarrow \mathcal{A}(C', X^{-n}) \xrightarrow{d^{-n} \circ -} \mathcal{A}(C', X^{-n+1}) \xrightarrow{d^{-n+1} \circ -} \dots \xrightarrow{d^{-1} \circ -} \mathcal{A}(C', X^0)$$

is exact.

Equivalently, an n -kernel of d^0 is such a sequence $(d^{-n}, d^{-n+1}, \dots, d^{-1})$, where d^{-k} is a weak cokernel of d^{-k+1} for all $1 \leq k \leq n-1$, and d^{-n} is a kernel of d^{-n+1} .

Dually, an n -cokernel of d^0 is a sequence

$$X^1 \xrightarrow{d^1} X^2 \xrightarrow{d^2} \dots \xrightarrow{d^n} X^{n+1}$$

such that d^k is a weak cokernel of d^{k-1} for $1 \leq k \leq n-1$ and d^n is a cokernel of d^{n-1} .

Remark 2.4 In the case $n = 1$, this indeed coincides with the usual definition of kernels and cokernels.

Now that we have a suitable notion of higher kernels and cokernels, we can consider the concept of n -exact sequences in the most logical way.

Definition 2.5 An n -exact sequence in $\mathcal{A}$ is a chain complex

$$X^0 \xrightarrow{d^0} X^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}$$

such that $(d^0, \dots, d^{n-1})$ is an n -kernel of d^n and $(d^1, \dots, d^n)$ is an n -cokernel of d^0 .

Remark 2.6 Once again, when $n = 1$, we get back to our usual notion of short exact sequence.

There are two final notions that need to be covered before proceeding with the definition of n -exact categories: that of n -pullback, n -pushout, and weak isomorphisms. Before covering these concepts, let us first remember what the cone of a morphism of complexes is:

Definition 2.7 Let X and Y be two complexes in $\text{Ch}(\mathcal{A})$ and $f : X \rightarrow Y$ be a morphism of complexes, or diagrammatically:

$$\begin{array}{ccccccc} X : & \cdots & \longrightarrow & X^{n-1} & \xrightarrow{d_X^{n-1}} & X^n & \xrightarrow{d_X^n} & X^{n+1} & \longrightarrow & \cdots \\ & & & \downarrow f^{n-1} & & \downarrow f^n & & \downarrow f^{n+1} & & \\ Y : & \cdots & \longrightarrow & Y^{n-1} & \xrightarrow{d_Y^{n-1}} & Y^n & \xrightarrow{d_Y^n} & Y^{n+1} & \longrightarrow & \cdots \end{array}$$

The *cone* of f , denoted $C(f)$, is the following complex

$$C(f) : \quad \cdots \longrightarrow X^n \oplus Y^{n-1} \xrightarrow{d_C^{n-1}} X^{n+1} \oplus Y^n \longrightarrow \cdots$$

where $d_C^{n-1} = \begin{pmatrix} -d_X^n & 0 \\ f^n & d_Y^{n-1} \end{pmatrix}$.

Before looking at the higher equivalents of pullbacks and pushouts, it is quite useful to examine the link that they have with the cone of a map.

Remark 2.8 Let $B^1 \xrightarrow{d_B^1} B^2$ and $A^2 \xrightarrow{f^2} B^2$ be two morphisms in $\mathcal{A}$ and consider the pullback of d_B^1 along f^2 :

$$\begin{array}{ccc} A^1 & \xrightarrow{d_A^1} & A^2 \\ f^1 \downarrow & \text{PB} & \downarrow f^2 \\ B^1 & \xrightarrow{d_B^1} & B^2 \end{array}$$

Clearly $f = (f^1, f^2)$ is a morphism of complexes since the square commutes. Thus we can consider its cone:

$$C(f) : \quad 0 \longrightarrow A^1 \xrightarrow{d_C^1} A^2 \oplus B^1 \xrightarrow{d_C^2} B^2 \longrightarrow 0$$

where $d_C^1 = \begin{pmatrix} -d_A^1 \\ f^1 \end{pmatrix}$, and $d_C^2 = \begin{pmatrix} f^2 & d_B^1 \end{pmatrix}$. We claim that d_C^1 is a kernel of d_C^2 . Indeed, consider a morphism $\begin{pmatrix} g \\ h \end{pmatrix} : C \rightarrow A^2 \oplus B^1$ such that the composition $d_C^2 \circ \begin{pmatrix} g \\ h \end{pmatrix} = 0$. The composition being 0 implies that $f^2 \circ g + d_B^1 \circ h = 0$, or $d_B^1 \circ h = -f^2 \circ g$. Consider the unique map $\hat{h} : C \rightarrow A^1$ given by $f^1 \circ \hat{h} = h$ and $d_A^1 \circ \hat{h} = -g$ (this map comes from the pullback property). We see that $d_C^1 \circ \hat{h} = \begin{pmatrix} -d_A^1 \circ \hat{h} \\ f^1 \circ \hat{h} \end{pmatrix} = \begin{pmatrix} g \\ h \end{pmatrix}$. Thus we have defined a unique map $\hat{h} : C \rightarrow A^1$ such that $d_C^1 \circ \hat{h} = \begin{pmatrix} g \\ h \end{pmatrix}$. That is, d_C^1 is a kernel of d_C^2 .

A dual result is true for pushouts, where d_C^2 is a cokernel of d_C^1 .

Basing ourselves on this observation, we can generalize the notion of pullback and pushout to higher degrees:

Definition 2.9 Let $X : X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^{n-1}} X^n$ be a complex in $\text{Ch}^n(\mathcal{A})$ and $f^0 : X^0 \rightarrow Y^0$ a morphism in $\mathcal{A}$. An n -pushout of X along f^0 is a morphism of complexes

$$\begin{array}{ccccccc} X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} & X^n \\ & \downarrow f^0 & & \downarrow f^1 & & & & \downarrow f^n \\ Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} & Y^n \end{array}$$

such that in the cone of f

$$C(f) : \quad X^0 \xrightarrow{d_C^{-1}} X^1 \oplus Y^0 \xrightarrow{d_C^0} \dots \xrightarrow{d_C^{n-1}} X^n \oplus Y^{n-1} \xrightarrow{d_C^n} Y^n$$

$(d_C^0, \dots, d_C^n)$ is an n -cokernel of d_C^{-1} .

The definition for the n -pullback of a chain complex is dual. That is, given a complex Y in $\text{Ch}^n(\mathcal{A})$ and a morphism $f^n : X^n \rightarrow Y^n$, an n -pullback of Y along f^n is a morphism of complexes $f : X \rightarrow Y$ such that in its cone $C = C(f)$, the sequence $(d_C^{-1}, \dots, d_C^{n-1})$ is an n -kernel of d_C^n .

The last notion required for defining n -exact categories is that of weak isomorphisms:

Definition 2.10 Let X and Y be two n -exact sequences in $\mathcal{A}$. We say $f : X \rightarrow Y$ is a *weak isomorphism* of complexes if there is $k \in \{0, 1, \dots, n\}$ such that f^k and f^{k+1} are isomorphisms.

When $n = 1$, we note that weak isomorphisms are isomorphisms of short exact sequences. This can be shown using the five lemma, which holds for any exact category (see [2, Lemma 8.9]) In the general case, however, weak isomorphisms are not necessarily isomorphisms, but a weaker result still holds:

Proposition 2.11 *Let X and Y be two n -exact sequences in $\mathcal{A}$ and let $f : X \rightarrow Y$ be a morphism of complexes. If f^k and f^{k+1} are isomorphisms for some $k \in \{0, 1, \dots, n\}$, then f induces an isomorphism in $\text{H}(\mathcal{A})$.*

The proof of this result can be found in [3, Proposition 2.7].

All of this preliminary work being done, we can finally proceed to define n -exact categories.

2.2 n -exact categories

With the proper tools put in place, it is quite straightforward to expand the definition of exact categories to longer exact sequences. Similar to what was done in chapter 1, we proceed with admissible n -exact sequences, admissible monics, and admissible epics:

Definition 2.12 Let $\mathcal{E}$ be a class of n -exact sequences in $\mathcal{A}$. An $\mathcal{E}$ -admissible n -exact sequence is an n -exact sequence that is in $\mathcal{E}$. An $\mathcal{E}$ -admissible monic is a morphism $d_X^0 : X^0 \rightarrow X^1$ such that there is an $\mathcal{E}$ -admissible n -exact sequence

$$X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^n} X^{n+1}.$$

Dually, an $\mathcal{E}$ -admissible epic is a morphism $d_X^n : X^n \rightarrow X^{n+1}$ such that there is an $\mathcal{E}$ -admissible n -exact sequence

$$X^0 \xrightarrow{d_X^0} \dots \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} X^{n+1}.$$

Whenever the context is clear, we will simply say admissible n -exact sequence (respectively admissible monic, admissible epic). We shall write $X^0 \xrightarrow{d_X^0} X^1$ for admissible monics and $X^n \xrightarrow{d_X^n} X^{n+1}$ for admissible epics.

The following definition simply adapts the axioms of exact categories to the definitions given in the previous section:

Definition 2.13 Let $\mathcal{E}$ be a class of n -exact sequences on $\mathcal{A}$. $\mathcal{E}$ is said to be an n -exact structure on $\mathcal{A}$ if it is closed under weak isomorphisms and it satisfies the following axioms:

- [E0] The sequence $0 \rightarrow 0 \rightarrow \dots \rightarrow 0$ is an $\mathcal{E}$ -admissible n -exact sequence;
- [E1] The composition of two $\mathcal{E}$ -admissible monics yields an $\mathcal{E}$ -admissible monic;

[E1^{op}] The composition of two $\mathcal{E}$ -admissible epics yields an $\mathcal{E}$ -admissible epic;

[E2] If $X : X^0 \xrightarrow{d_X^0} \dots \xrightarrow{d_X^n} X^{n+1}$ is an $\mathcal{E}$ -admissible n -exact sequence and $X^0 \xrightarrow{f^0} Y^0$ is a morphism in $\mathcal{A}$, there exists an n -pushout of $(d_X^0, \dots, d_X^{n-1})$ along f^0

$$\begin{array}{ccccccc} X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} & X^n & \xrightarrow{d_X^n} & X^{n+1} \\ & \downarrow f & & \downarrow f^0 & & \downarrow f^1 & & \downarrow f^n & & \\ Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} & Y^n & & \end{array}$$

such that d_Y^0 is an $\mathcal{E}$ -admissible monic;

[E2^{op}] If $Y : Y^0 \xrightarrow{d_Y^0} \dots \xrightarrow{d_Y^n} Y^{n+1}$ is an $\mathcal{E}$ -admissible n -exact sequence and $X^{n+1} \xrightarrow{f^{n+1}} Y^{n+1}$ is a morphism in $\mathcal{A}$, there is an n -pullback of $(d_Y^1, \dots, d_Y^n)$ along f^{n+1}

$$\begin{array}{ccccccc} X : & & & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} & X^n & \xrightarrow{d_X^n} & X^{n+1} \\ & & & \downarrow f^1 & & & & \downarrow f^n & & \downarrow f^{n+1} \\ & \downarrow f & & & & & & & & \\ Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} & Y^n & \xrightarrow{d_Y^n} & Y^{n+1} \end{array}$$

such that d_X^n is an $\mathcal{E}$ -admissible epic.

Remarks 2.14 • The axiom [E0] given here seems to be less restrictive than the [E0] and [E0^{op}] of exact categories. However, if $n \geq 2$, note that identity morphisms are also admissible monics since

$$\begin{array}{ccccccc} X & \xrightarrow{1_X} & X & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & & & \downarrow \\ 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \end{array}$$

is a weak isomorphism. Identity morphisms are admissible epics for the same reason.

- One can show that n -kernels and n -cokernels are unique up

to weak isomorphism. As such, all n -kernels (respectively n -cokernels) of an admissible epic (respectively admissible monic) fit into an admissible n -exact sequence.

2.3 n -cluster tilting subcategories

Similar to extension-closed subcategories of abelian categories, there is a higher degree notion that gives us natural examples of n -exact categories. But this requires a few more definitions, namely what being functorially finite means.

Definition 2.15 Let $\mathcal{A}$ be an additive category and $\mathcal{D}$ be a full subcategory of $\mathcal{A}$. $\mathcal{D}$ is called *covariantly finite* if for any object $A \in \mathcal{A}$ there exists an object $L_A \in \mathcal{D}$ and a morphism $f_A : A \rightarrow L_A$ such that for any morphism $g : A \rightarrow D$ where $D \in \mathcal{D}$, there exists a morphism $h : L_A \rightarrow D$ such that $g = h \circ f_A$.

Dually, $\mathcal{D}$ is called *contravariantly finite* if for any object $A \in \mathcal{A}$ there exists an object $R_A \in \mathcal{D}$ and a morphism $f_A : R_A \rightarrow A$ such that for any morphism $g : D \rightarrow A$ where $D \in \mathcal{D}$, there exists a morphism $h : D \rightarrow R_A$ such that $g = f_A \circ h$.

Both properties can be summarized in the following diagrams:

$$\begin{array}{ccc}
 A & \xrightarrow{f_A} & L_A \\
 & \searrow h & \downarrow \exists g \\
 & & D
 \end{array}
 \qquad
 \begin{array}{ccc}
 R_A & \xrightarrow{f_A} & A \\
 \uparrow \exists g & \nearrow h & \\
 D & &
 \end{array}$$

covariantly finite
contravariantly finite

$\mathcal{D}$ is then said to be *functorially finite* if it is both covariantly and contravariantly finite.

Remark 2.16 In the context of module categories, this is very often true for any subcategory. In fact, if A is an Artin algebra, one can show

that every subcategory of $\text{mod } A$ is functorially finite if and only if A is of finite representation type (see [1, Proposition 1.2]).

Definition 2.17 Let $\mathcal{A}$ be an abelian category and $\mathcal{D}$ be a full subcategory of $\mathcal{A}$. $\mathcal{D}$ is an *n -cluster tilting category* if

- it is functorially finite;
- $\mathcal{D} = \{C \in \mathcal{A} \mid \text{Ext}_{\mathcal{A}}^i(C, D) = 0 \ \forall D \in \mathcal{D}\};$
- $\mathcal{D} = \{C \in \mathcal{A} \mid \text{Ext}_{\mathcal{A}}^i(D, C) = 0 \ \forall D \in \mathcal{D}\}.$

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